Name_____ Date_____ Period_____

LESSON 5-7 · HOMEWORK PACKET · SHEET 1 OF 2

Surface Area of Rectangular Prisms

Maryland Standard 6.G.A.4 · Learning target – I can find the surface area of a rectangular prism by adding the areas of its six faces.

REMEMBER

- **Surface area** is the total area of all the faces, in **square units**.
- A rectangular prism has 3 pairs of matching faces: **SA = 2lw + 2lh + 2wh**.

WORKED EXAMPLE

A box is 8 in × 3 in × 2 in. Find its surface area.

Step 1 One face from each pair: 8 × 3 = 24, 8 × 2 = 16, 3 × 2 = 6

Step 2 Add: 24 + 16 + 6 = 46

Step 3 Each face has a twin, so double: 2 × 46 = 92

Answer **92 in²**

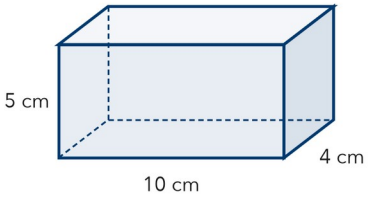
PRACTICE

show your work · label units

1. Find the surface area of the prism.

Show your work.

Answer_____ cm²



2. Find the surface area of each solid.

a. cube, edges 3 cm _____ cm²

b. cube, edges 10 ft _____ ft²

c. prism 4 m × 2 m × 1 m _____ m²

d. prism 5 in × 5 in × 2 in _____ in²

e. cube, edges 5 in _____ in²

f. prism 6 cm × 3 cm × 2 cm _____ cm²

3. **Gift Box.** Nadia wraps a box that is 12 in long, 8 in wide, and 4 in tall. Her paper covers 400 in². Is that enough to cover the box? Explain.

Show your work.

Answer_____

4. **Toy Chest.** A toy chest is 4 ft long, 2 ft wide, and 3 ft tall. Leo paints all faces but the bottom. How many ft² does he paint?

Answer_____ ft²

5. **Find the Error.** Diego says the surface area of a cube with 2 cm edges is 8 cm² because 2 × 2 × 2 = 8. Explain his mistake and find the surface area.

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME Pick a box at home. Point to each pair of matching faces and tell someone how you would find its surface area.

Name_____ Date_____ Period_____

LESSON 5-7 · HOMEWORK PACKET · SHEET 2 OF 2

Surface Area of Triangular Prisms

Maryland Standard 6.G.A.4 · Learning target – I can find the surface area of a triangular prism using its net.

REMEMBER

- A triangular prism has **2 triangle bases** and **3 rectangles**.
- Each rectangle is one side of the triangle × the prism's length. Add all **5 faces**.

WORKED EXAMPLE

Prism 10 cm long. Ends are right triangles with sides 3, 4, 5 cm.

Step 1 Triangles: 2 × (½ × 3 × 4) = 12

Step 2 Rectangles: 30 + 40 + 50 = 120

Step 3 Add: 12 + 120 = 132

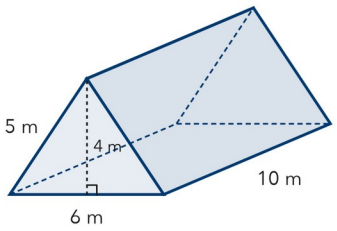
Answer **132 cm²**

PRACTICE

show your work · label units

1. Find the surface area. The triangle ends have sides 5 m, 5 m, and 6 m.

Show your work.



Answer_____ m²

2. Find each area.

a. 2 triangles, base 8 ft, height 3 ft: _____ ft²

b. 1 rectangle, 5 ft by 9 ft: _____ ft²

3. Tent. A tent is a triangular prism 9 ft long. Each end is a triangle with base 8 ft, height 3 ft, and two 5 ft sides. Find the fabric needed for all 5 faces, including the floor.

Answer_____ ft²

4. Doorstop. A wooden doorstop is a triangular prism 4 in long. Its ends are right triangles with sides 5 in, 12 in, and 13 in. Find its surface area.

Answer_____ in²

5. Find the Error. For the prism in Exercise 1, Lucas found the rectangles as 6 × 10, 4 × 10, and 4 × 10. Explain his mistake.

_____

_____

TEST PRACTICE

choose the best answer

6. Triangle ends: 9 cm² each. Rectangles: 20, 20, and 24 cm². What is the surface area?

A 64 cm² **B** 73 cm² **C** 82 cm² **D** 146 cm²

7. A prism is 8 cm long. Its triangle ends have sides 6, 8, and 10 cm. What is the total area of the 3 rectangles?

A 24 cm² **B** 192 cm² **C** 240 cm² **D** 480 cm²

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___